BEYOND EARTH – SPACE EXPLORATION AND ASTRONOMY WEBSITE

Overview

The "Beyond Earth" website is an educational web application developed to provide users with information about space exploration, astronomy, celestial events, constellations, astrophotography, and recent developments in space science. The website serves as a learning platform that presents astronomical concepts in a visually appealing and interactive manner.

Objective

The objective of this project is to develop an educational website that provides users with information about astronomy, celestial events, constellations, astrophotography and current developments in space exploration.

Live website- https://beyond-earth-nine.vercel.app/

Features

- Responsive navigation bar
- Dedicated pages for astronomy topics
- Celestial Events section
- Constellations section
- Astrophotography section
- Space News section
- Interactive clickable cards
- Image-based content presentation
- Responsive design for different screen sizes

Technology Stack

Frontend:
- HTML5
- CSS3
- JavaScript

Development Tools:
- Visual Studio Code

• GitHub
• GitHub Pages

Website Structure

Pages

1. Home Page (index.html)

Acts as the central page of the website and provides:

- Introduction to Space Exploration and Astronomy
- Navigation menu for all sections
- Featured images
- Quick access cards to topic pages

2. Celestial Events Page (box1.html)

Handles information related to astronomical events such as:

- Solar Eclipses
- Lunar Eclipses
- Meteor Showers
- Supernova Explosions
- Gamma-Ray Bursts
- Auroras

3. Constellations Page (box2.html)

Provides educational content regarding:

- Orion
- Ursa Major
- Ursa Minor
- Cassiopeia
- Leo
- Scorpius
- Cygnus

It also includes guidelines for observing constellations in the night sky.

4. Astrophotography Page (box3.html)

Contains information about:

- Wide-Field Astrophotography
- Planetary Photography
- Deep-Sky Photography
- Time-Lapse Photography

- Aurora Photography

5. Space News Page (right-image-page.html)

Displays recent updates related to:

- NASA Missions
- ISRO Missions
- Artemis Program
- Mars Exploration
- Satellite Launches

Styling

Style.css

The CSS file is responsible for:

- Responsive page layouts
- Navigation bar design
- Content box styling
- Image presentation
- Hover effects and animations
- Mobile device compatibility

JavaScript Functionality

script.js

Contains functions responsible for:

Navigation Management

- Redirecting users between pages
- Handling clickable content boxes
- Improving user interaction and website usability

User Interface Components

1.Navigation Bar

Provides access to:

- Home
- Space News
- Celestial Events
- Constellations
- Astrophotography

2.Content Sections

Each page contains:

- Educational text
- Related images
- Structured headings and lists
- Responsive layouts

3.Footer

Displays copyright information and website branding.

Screenshots

Figure 1. Home Page

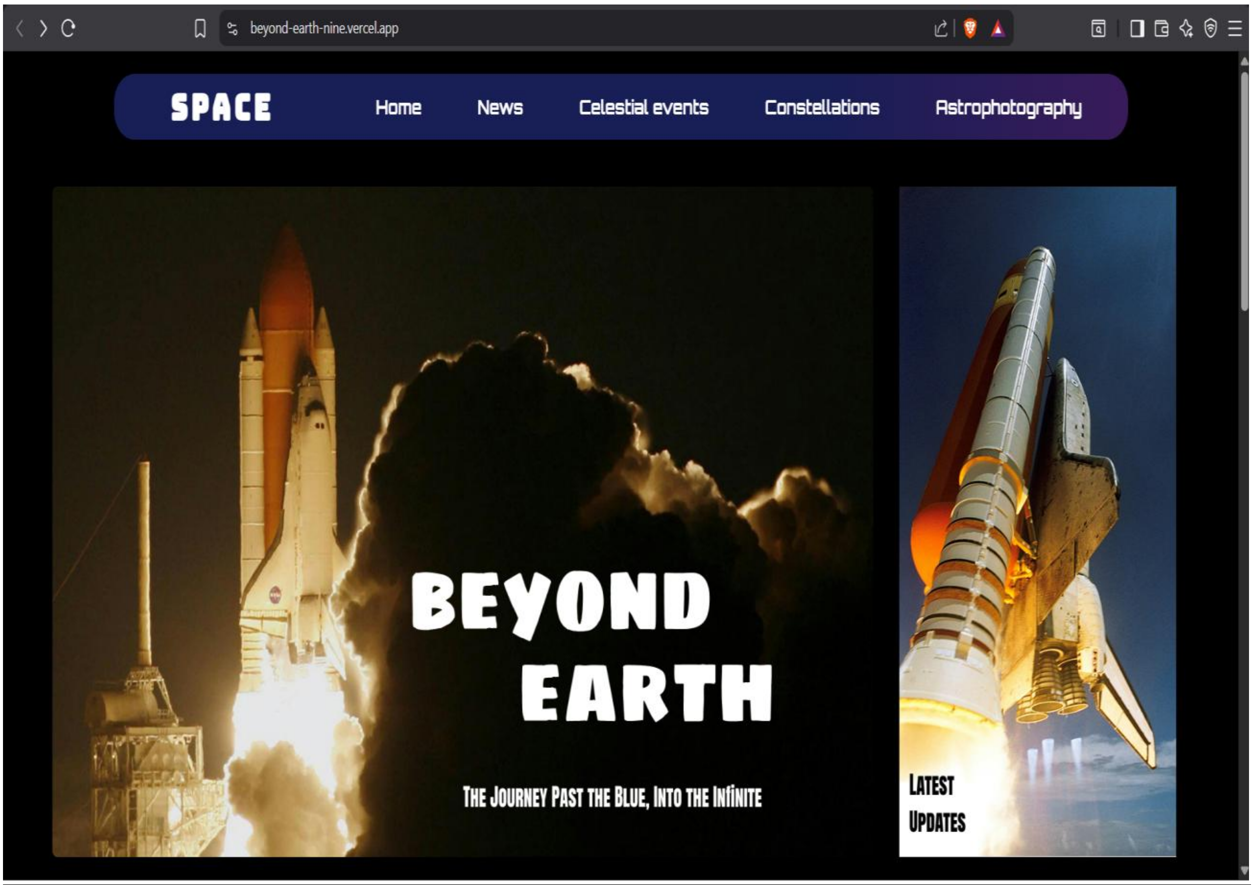


Figure 2. Mid-Section

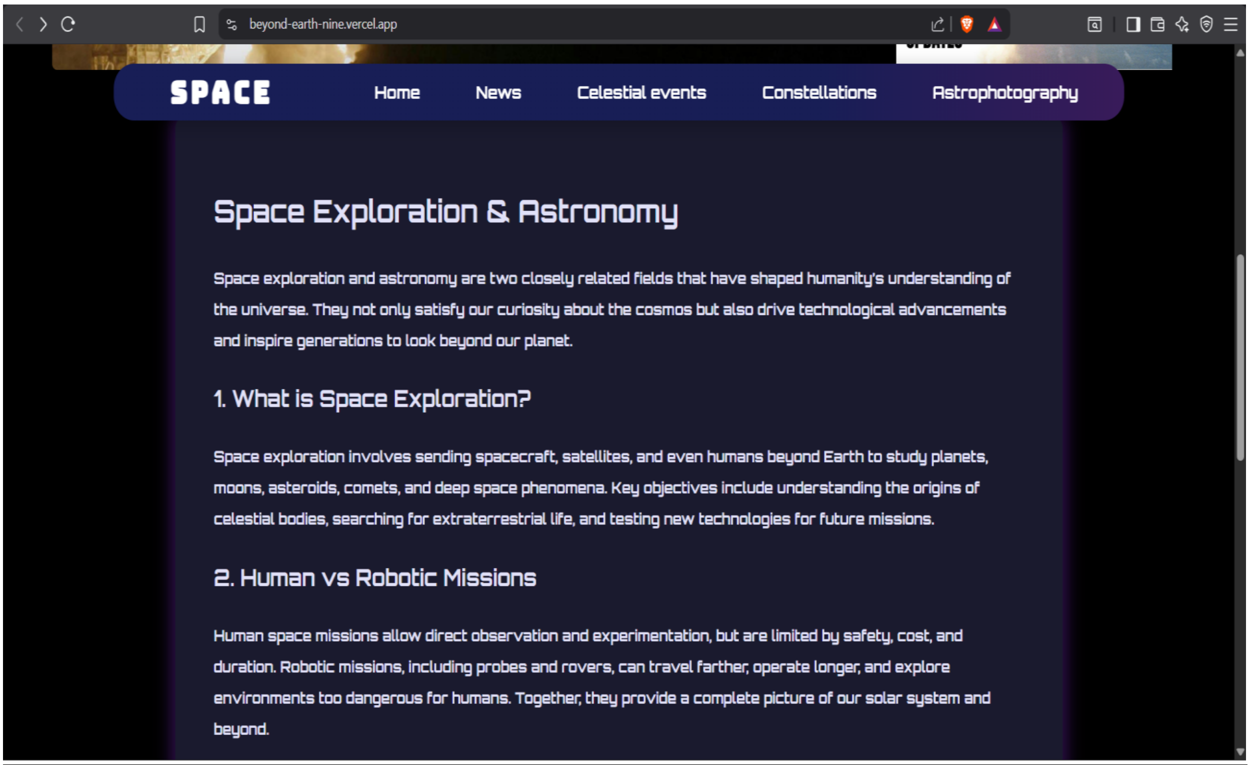


Figure 3.Bottom of the page with cards that redirect to their respective pages when clicked

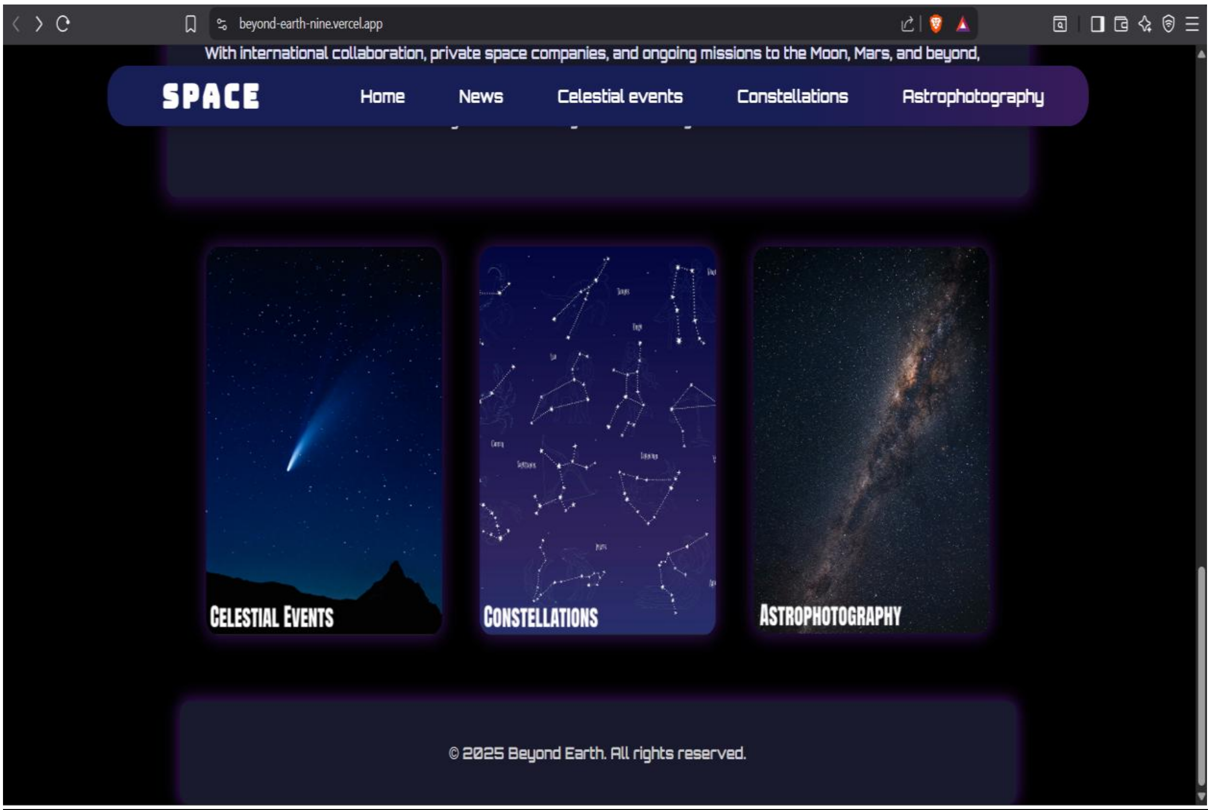


Figure 4.Space news page

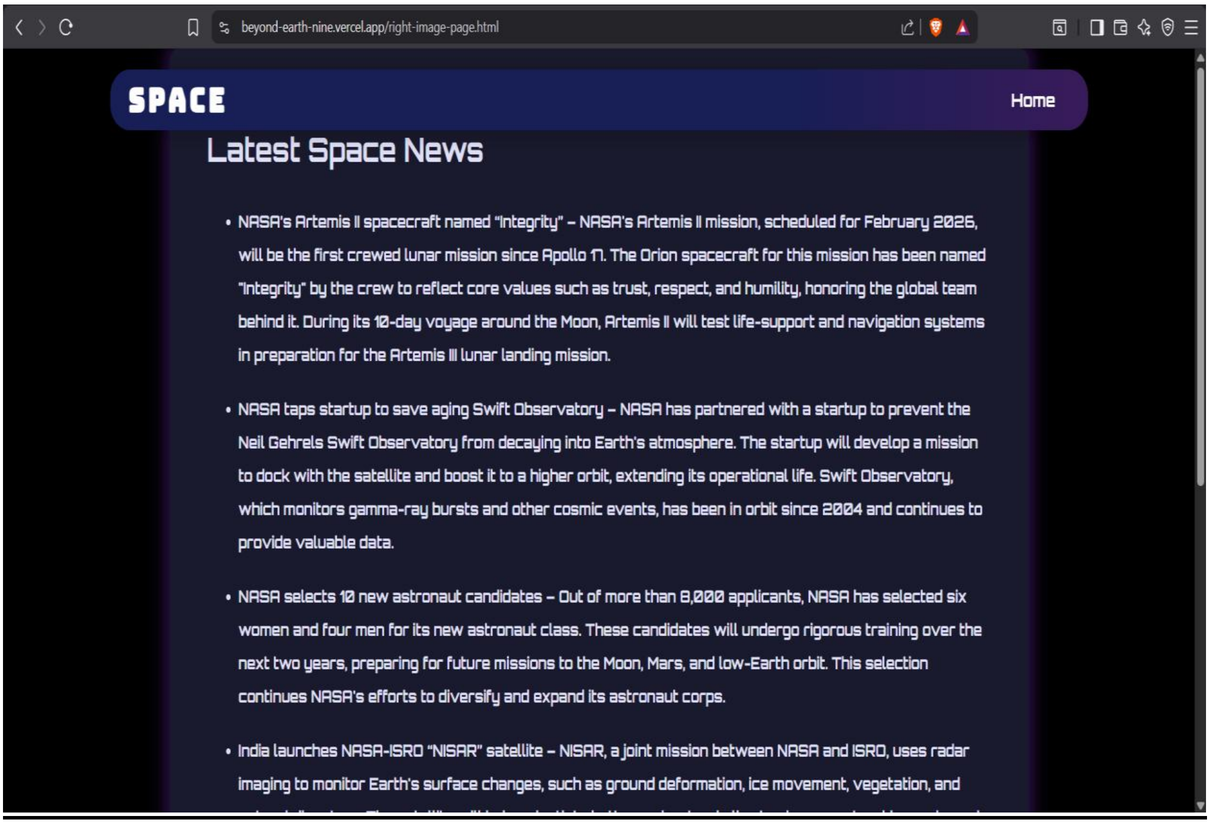


Figure 5.Celestial events page

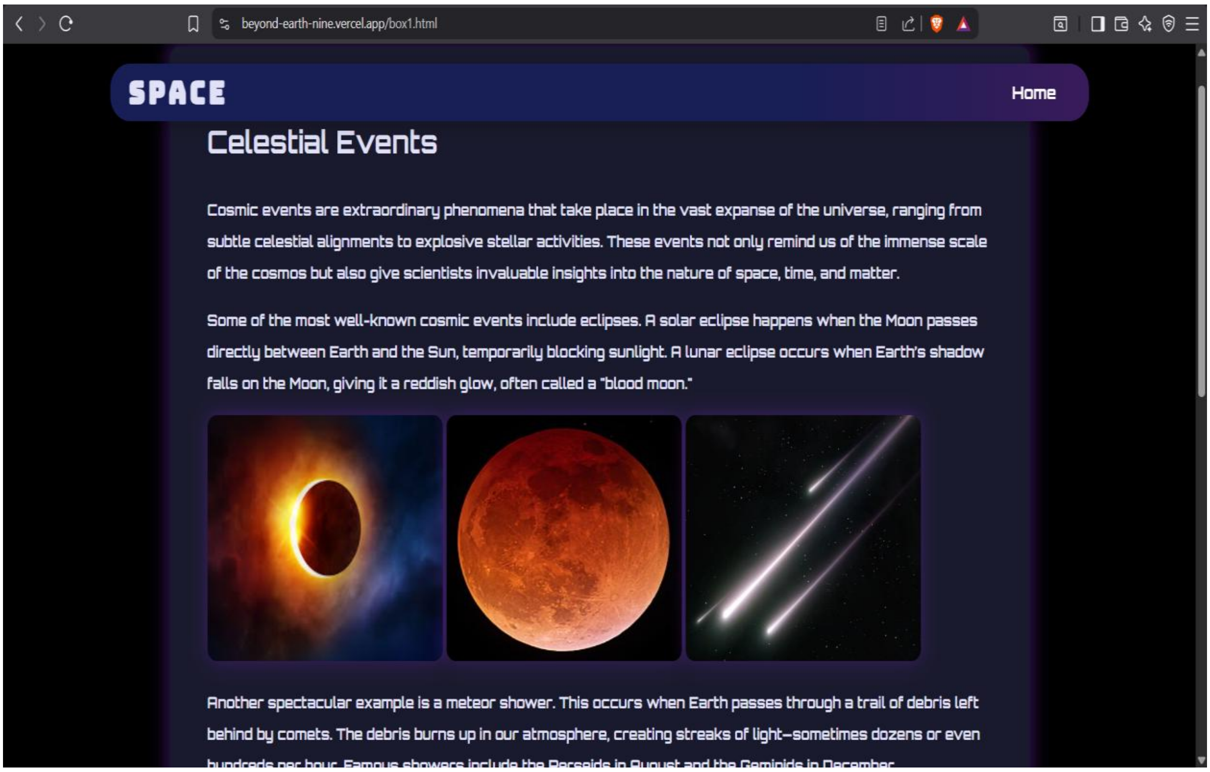


Figure 6.Constellations page

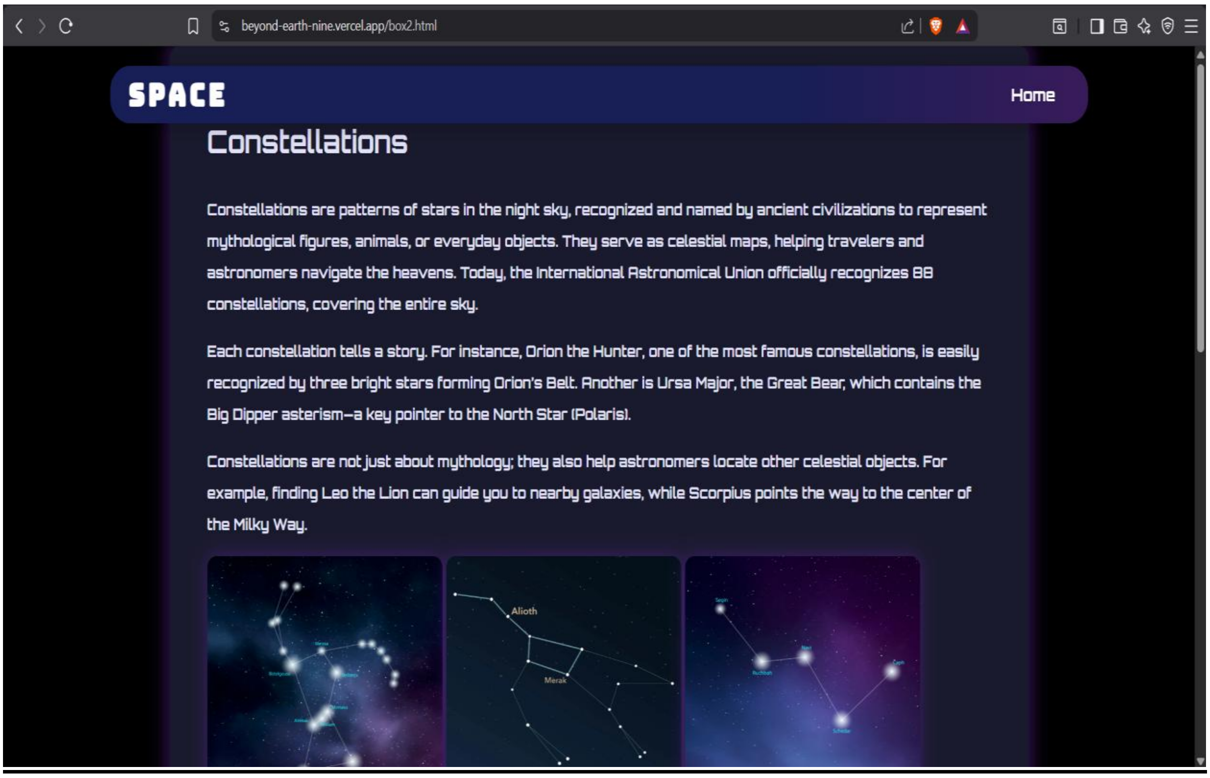
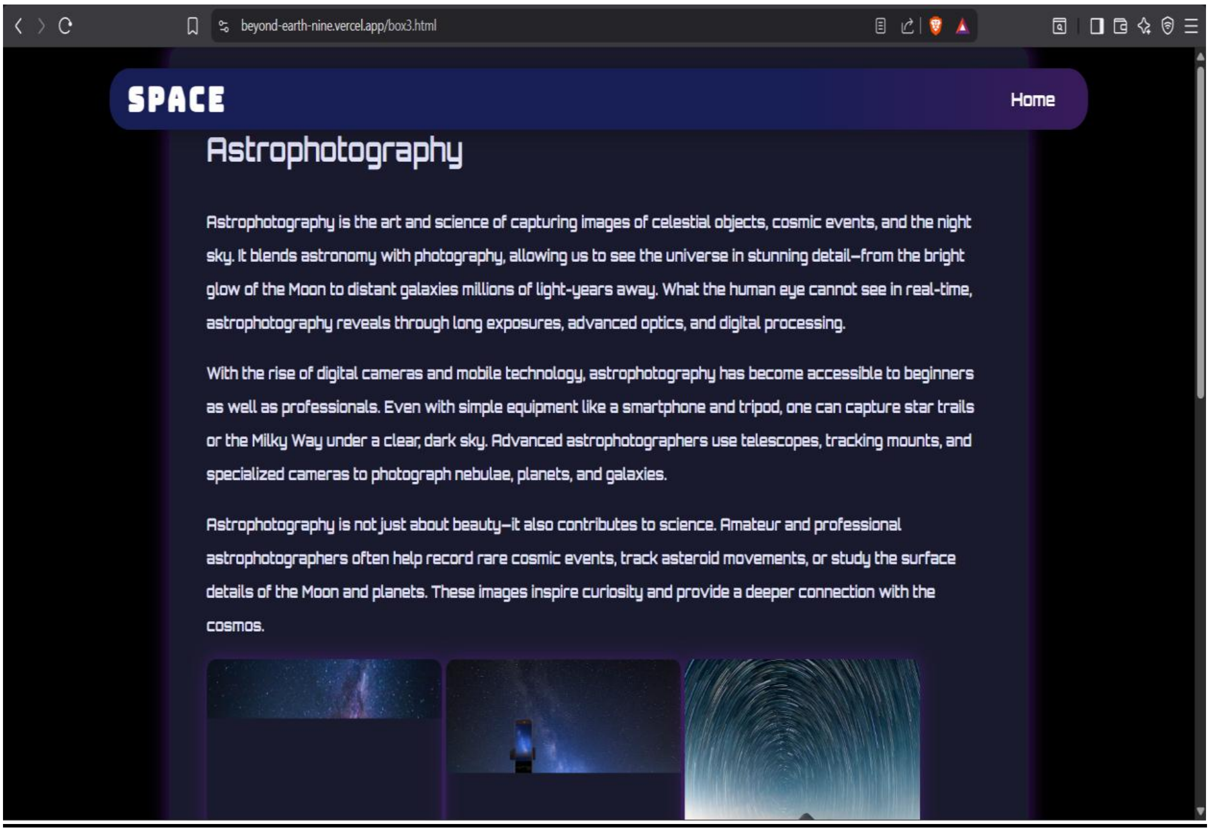


Figure 7.Astrophotography page



System Architecture

```
User
   |
Home Page
   |
   |---Celestial Events
   |--- Constellations
   |--- Astrophotography
   |---Space News
   |
Content Display
```

Future Enhancements

- Search functionality
- Dark/Light theme toggle
- User accounts and bookmarks
- Interactive star maps
- Astronomy quiz module
- Real-time NASA/ISRO API integration
- Celestial event calendar

Conclusion

The Beyond Earth website successfully provides educational content related to astronomy and space exploration through a simple, responsive, and interactive web interface. The project demonstrates the practical application of HTML, CSS, and JavaScript in developing a multi-page informational website.